

Zongmin Zhang

HKUST Computer Science Undergraduate | AI for Science Researcher

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Website Scholar ORCID GitHub LinkedIn X

EDUCATION

The Hong Kong University of Science and Technology (HKUST) 09/2023 – Present

BEng in Computer Science (Minor in Chemistry), Department of Computer Science and Engineering (CSE), School of Engineering (SENG)

CGA: 3.999/4.300 (top 2%) | Major CGA (MCGA): 4.094/4.300 | Dean's List ×5 | HKSAR Government Scholarship ×3

Selected A+ Coursework:

Chemistry: General Chemistry I & II, Organic Chemistry, Undergraduate Research.

CS & AI: OOP (C++/Java), Data Structures, Algorithms, Game Programming.

Math & Physics: Multivariable Calculus, Matrix Algebra, Statistics, Probability, Quantum Information.

PUBLICATIONS

[1] Yuqi Fang, **Zongmin Zhang**, Lixue Cheng, Yong Huang. "Bayesian Optimization–Assisted Human–Machine Collaborative Design of ALK L1196M/G1202R Inhibitors." *In preparation*, 2026.

[2] **Zongmin Zhang**, Jingtong Wu, Shaokang Li, Lixue Cheng. "SFMSafetyLens: A Multi-Scale Diagnostic Framework for the Safety Spectrum of Scientific Foundation Models." *In preparation*, 2026.

[3] **Zongmin Zhang**, Kexiang Hao, Xinyi Liu, Lixue Cheng. "ChemTriage: Matched Records Test How Language Models Turn Chemical Evidence into Decisions." *In preparation*, 2026.

[4] **Zongmin Zhang**, Yuyang Lou, Bowen Zhang, Junwu Chen, Ryo Kuroki, Xuan Vu Nguyen, Edvin Fako, Lixue Cheng, and Philippe Schwaller. "AdsMind: A Physics-Grounded Multi-Agent System for Self-Correcting Discovery of Adsorption Configurations on Heterogeneous Catalyst Surfaces." *arXiv preprint* arXiv:2606.19152, 2026. Submitted to **Machine Learning: Science and Technology** (Under Review).

[5] Zhilong Song, **Zongmin Zhang**, and Lixue Cheng. "Autonomous Heterogeneous Catalyst Discovery with a Self-Evolving Multi-Agent Digital Twin." *arXiv preprint* arXiv:2606.05050, 2026. Submitted to **Nature Communications** (Major Revision Underway).

[6] Aritra Roy et al. (including **Zongmin Zhang**). "From Knowledge to Action: Outcomes of the 2025 Large Language Model (LLM) Hackathon for Applications in Materials Science and Chemistry." *arXiv preprint* arXiv:2605.03205, 2026.

RESEARCH EXPERIENCE

Undergraduate Research Assistant (UGRA), AI for Physical Sciences (AI4PhysSci) Lab, Department of Chemistry, School of Science, HKUST 02/2026 – Present

Supervisor: Prof. Lixue Cheng

- Conducting AI for Science research on chemistry-oriented machine learning, multi-agent systems (MAS)

and safety-aware autonomous scientific discovery, leading to preprints on AdsMind, CatDT, and ChemTriage (in prep.).

Developments and Applications of Orbital-Based Learning as a General and Accurate Property Predictor 02/2026 – 05/2026

HKUST Undergraduate Research Opportunities Program (UROP) | Supervisor: Prof. Lixue Cheng

- Investigated orbital-based machine learning representations and benchmarked baseline architectures to build general and accurate property predictors for chemical systems.

Seeing Surfaces Differently: Agentic Simulation of Atomic Rearrangement and Surface Reactivity 09/2025 – 01/2026

Semester Project, Laboratory of Artificial Chemical Intelligence (LIAC), ISIC, EPFL | Supervisors: Prof. Philippe Schwaller & Dr. Edvin Fako

- Developed a modular framework integrating LLM reasoning backends (e.g., Gemini, OpenRouter, Ollama) with atomistic simulation engines (e.g., MACE, OpenMD).
- Built automated conformer placement and relaxation toolchains, and curated a 20-surface benchmark dataset spanning bimetallic alloys and metal oxides (e.g., Mo_3Pd, CuZnO), establishing the prototype for AdsMind.

Assess User Experience to Design Effective Visual Representation and Interaction in Virtual Reality 02/2025 – 05/2025

HKUST Undergraduate Research Opportunities Program (UROP) | Supervisor: Prof. Xiaojuan Ma

- Developed a multi-user collaborative VR ideation environment in Unity 6 (URP) for Meta Quest 3, resolving legacy-to-URP shader incompatibilities and texture bindings.
- Implemented real-time bimanual grab, release, and wall auto-attachment using Meta XR Interaction SDK and custom C# scripts, synchronized across headsets via Unity Netcode for GameObjects (NGO).

Knowledge Discovery over Databases 06/2024 – 12/2024

HKUST Undergraduate Research Opportunities Program (UROP) | Supervisor: Prof. Raymond Chi-Wing Wong

- Implemented a tensor-level PyTorch data cache using vector search utilities to eliminate redundant latent vector re-evaluations for recurring item IDs, significantly reducing deep neural network (DNN) model training overhead.
- Trained an MLP autoencoder to compress high-dimensional embedding vectors from 128D to 8D with a cosine similarity of $\sim$0.9 and integrated warm-up and decay learning rate schedulers to stabilize optimization.

HONORS & AWARDS

HKSAR Government Scholarship Fund – Scholarship for Outstanding Performance AY 2024/25 – 2026/27
(Awarded for 3 consecutive years, HK$80,000 p.a.)

Hong Kong, China–Asia-Pacific Economic Cooperation Scholarship (Awarded for 3 consecutive years) AY 2024/25 – 2026/27

HKUST Study Abroad Funding Support (HK$10,000)

- **HKUST Study Abroad Grant** Fall 2025
- **HKSAR Government Scholarship Fund – Reaching Out Award (ROA)** AY 2025/26

HKUST School of Engineering (SENG) Dean's List (Awarded for all terms at HKUST) Fall 2023 – Spring 2026

EXCHANGE & VISITING

HKUST SENG Undergraduate Outbound Exchange Student 09/2025 – 02/2026

École Polytechnique Fédérale de Lausanne (EPFL), School of Computer and Communication Sciences (IC), Lausanne, Switzerland

Association of Pacific Rim Universities (APRU) Virtual Student Exchange (VSE) Program 02/2025 – 06/2025

Shanghai Jiao Tong University (SJTU), School of Materials Science and Engineering (MSE), Shanghai, China
Relevant coursework: MSE2602 Materials Chemistry.

INDUSTRY EXPERIENCE

Computer Vision Algorithm Intern 06/2025 – 08/2025

Sansen Well (Shanghai) Robotics Co., Ltd., Shanghai, China

- Independently built an end-to-end multi-view 3D reconstruction pipeline in C++ and OpenCV for industrial wire-rope inspection.
- Implemented arcsin cylindrical unwrapping, dynamic programming seam-finding with Laplacian blending for seamless 360° texture stitching, and VTK-based 3D mesh rendering.

U3D VR Developer Intern 12/2024 – 01/2025

Qingdao Virtual Reality Institute Co., Ltd., Qingdao, Shandong, China

Frontend Developer Intern 08/2024

Lucent Qingdao R&D Center, Qingdao, Shandong, China

TEACHING & SERVICE

IOP Trusted Reviewer, *Machine Learning: Science and Technology (MLST) & Machine Learning: Engineering (MLENG)* 2026

Reviewer, ICML 2026 AI for Science Workshop & NeurIPS 2026 AI for Science Workshop (OpenReview) 2026

HKUST CSE Programming Commons Undergraduate Teaching Assistant (UGTA) 2026, Fall 2024

HKUST COMP2011 Undergraduate Teaching Assistant (UGTA) Spring 2025

Student Representative, HKUST SENG Mainland Undergraduate Recruitment Trip – JEE Main Round Interview Mingling Sessions, Guangzhou, China 06/2026 – 07/2026

COMMUNITY & LEADERSHIP

Co-Founder & Vice President (VP), HKUST AI for Chemistry Club 09/2026 – Present

(Department-Associate Group (DAG) under the Department of Chemistry)

- Initiated a student-led academic community for seminars, reading groups, workshops, and peer learning at the intersection of artificial intelligence and chemistry.

Member & Participant, Hong Kong Tsingtao Association (HKQDA) & Mainland Students and Scholars Society, UG Section (MSSSUG), HKUST 09/2023 – Present

Member & Participant, Chinese Students and Scholars Association in Lausanne (CSSA Lausanne) & Crazy Hikers 09/2025 – 02/2026

First Prize, Outstanding Volunteer & Data Analysis Professional Skill Certificate, 5th National University Students' Data Analysis Popular Science Knowledge Competition (Theory Track) 05/2026

First Prize & Outstanding Publicity Ambassador, 2026 National University Student Science and Technology Knowledge Competition 05/2026

First Prize & Volunteer, 5th National Students' Science Literacy Knowledge Popularization Activity 05/2026

Finalist Award (Team Category) & **Team Leader**, HKUST GenAI Hackathon Competition on a Putonghua Web Tool (Prize: HK$800 per team) [Project: PSC-Copilot] 03/2026

Team Leader, Open Scientific Intelligence (OSI) Hackathon [Project: SciExchange] 10/2026

SKILLS & MISCELLANEOUS

Technical Stack:

Languages & Systems: Python (proficient), C/C++ (proficient), Java (proficient), Rust, C#, Scala, R, JavaScript, HTML, LaTeX, Markdown, Linux/Shell scripting, OS internals, kernel-level programming

Deep Learning & ML: PyTorch, TensorFlow, CUDA, TensorRT, NumPy, Keras, scikit-learn

AI Agents & LLMs: LangGraph, LangChain, prompt engineering, RAG, Model Context Protocol (MCP), agent tool & skill design (e.g., Codex and Claude Code), harness engineering

AI4Science & Cheminformatics: RDKit, Atomic Simulation Environment (ASE), MACE-MP

Vision, Graphics & XR: OpenCV, YOLO, SAM, 3D Reconstruction, Visualization Toolkit (VTK), Unity, Meta XR SDK

Language Proficiency: Mandarin (Native), English (Full professional proficiency; Gaokao English: 142/150), Japanese (Limited working proficiency; approx. JLPT N3 level), French (Elementary proficiency; CEFR A1 level), Cantonese (Elementary proficiency)

Professional Affiliations: Student Member of ACS, RSC, CCS (Chemistry); IEEE, ACM, CCF, HKCS (Computer Science) 2026 – Present

Research Ethics: Responsible Conduct of Research (UROP), CITI Program, HKUST Valid until 06/2029